

Cost-Adjusted Bayesian Model Selection in the Presence of Factors

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Dissertation

Master

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Introduction





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Literature Review



Exponential Dispersion Model Family

- It's a family of distributions. A random variable $y \sim \text{EDM}(\mu, a(\phi))$ belongs to it if (canonical form):

$$f(y \mid \theta, \phi) = \exp \left[\frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi) \right]. \quad (1)$$

- $\mathbb{E}(y) = \mu = b'(\theta)$ (one-to-one relationship between the mean μ and θ);
- $\mathbb{V}(y) = a(\phi)b''(\theta) = a(\phi)V(\mu) > 0$ (the *variance function* uniquely identifies the distribution within the EDM family).

Generalized Linear Models

- **Random Component:** for each observation in $\mathbf{Y} = (y_1, \dots, y_n)^\top$, $y_i \stackrel{\text{indep}}{\sim} \text{EDM}(\mu_i, a(\phi)), i = 1, \dots, n$.
- **Systematic Component:** the response means, $\mathbb{E}[\mathbf{Y}] = \boldsymbol{\mu}$, are related to a linear combination of the explanatory variables, $\boldsymbol{\eta} = (\eta_1, \dots, \eta_n)^\top$, through the link function,

$$\eta_i = h(\mu_i), \quad (2)$$

with

$$\boldsymbol{\eta} = \mathbf{1}_n \alpha + \mathbf{X} \boldsymbol{\beta}, \quad (3)$$

where $\mathbf{X} = [\mathbf{x}_1 \mid \dots \mid \mathbf{x}_k]$ is the design matrix; $\mathbf{x}_j = (x_{1j}, \dots, x_{nj})^\top$ is the n -dimensional vector of observations for the j -th explanatory variable, $j = 1, \dots, k$; and β_j is the corresponding regression coefficient, so that $\boldsymbol{\beta} = (\beta_1, \dots, \beta_k)^\top$.

The Variable Selection Problem

What subset of $\mathbf{x}_1, \dots, \mathbf{x}_k$ "best" explains $\mathbf{Y}$?

- **(Point) Estimation Methods:** a single model is fit; computationally cheap. How to properly induce *sparsity*? $\mathcal{M}_F$ most likely contains many variables with uncertain effects on $\mathbf{Y}$;
- **Model Selection Methods:** considers all 2^k models in $\mathcal{M}$ (main effects of numerical variables only); enables model-averaged estimates. Promotes sparsity naturally. Overall Type I error increases with k . How to correct for the *multiple comparisons problem*?

Solutions for both approaches are offered under the *Frequentist* and *Bayesian* frameworks. Distinctions in the treatment of model parameters and methods for quantifying uncertainty.



Bayesian Model Selection

- Probabilistic framework that accounts for both model and parameter uncertainty. All models in $\mathcal{M}$ are from the same EDM family.
- Competing models differ only in the subset of explanatory variables included in (3). Model uncertainty $\rightarrow \boldsymbol{\gamma} = (\gamma_1, \dots, \gamma_k)^\top$, a binary vector in $\{0, 1\}^k$ encoding the predictors included in a given model ($\gamma_j = 1 \iff \mathbf{x}_j$ is included):

$$(y_i \mid \mu_i(\boldsymbol{\gamma}), \phi) \stackrel{\text{indep}}{\sim} \text{EDM}\left(\mu_i(\boldsymbol{\gamma}), \frac{\phi}{w_i}\right), \quad i = 1, \dots, n; \quad (4)$$

$$\boldsymbol{\eta}_{\boldsymbol{\gamma}} = \mathbf{1}_n \alpha + \mathbf{X} \text{diag}(\boldsymbol{\gamma}) \boldsymbol{\beta} = \mathbf{1}_n \alpha + \mathbf{X}_{\boldsymbol{\gamma}} \boldsymbol{\beta}_{\boldsymbol{\gamma}}. \quad (5)$$

- each $\eta_i(\boldsymbol{\gamma})$ is related with $\mu_i(\boldsymbol{\gamma})$ through the link function; $\mathcal{M}_{\emptyset}$ includes only α ;
- $\mathbf{X}_{\boldsymbol{\gamma}}$: model-specific design matrix with $k_{\boldsymbol{\gamma}} = \sum_{j=1}^k \gamma_j \leq k$ columns (associated with the nonzero elements of $\boldsymbol{\gamma}$); $\boldsymbol{\beta}_{\boldsymbol{\gamma}}$ is a $k_{\boldsymbol{\gamma}}$ -length vector.

Bayesian Model Selection

Hierarchical mixture framework (with known link function):

- (Prior) Model Uncertainty $\rightarrow f(\mathcal{M}_\gamma)$ or $f(\gamma)$; models $\notin \mathcal{M}$ have 0 prior probability;
- (Prior) Parameter Uncertainty $\mid \mathcal{M}_\gamma \rightarrow f(\boldsymbol{\psi}_\gamma \mid \mathcal{M}_\gamma)$, with $\boldsymbol{\psi}_\gamma = (\alpha, \boldsymbol{\beta}_\gamma, \phi) \in \Psi_\gamma$;
- Sampling Model $\mid \mathcal{M}_\gamma, \boldsymbol{\psi}_\gamma \rightarrow f(\mathbf{Y} \mid \boldsymbol{\psi}_\gamma, \mathcal{M}_\gamma)$; see (4); joint p.m.f. / p.d.f. $\propto$ *likelihood*, $\mathcal{L}(\boldsymbol{\psi}_\gamma \mid \mathbf{Y}, \mathcal{M}_\gamma)$.

Prior information is updated based $\mathbf{Y}$ (fixed), through Bayes' theorem (Casella and R. L. Berger 2002). Posterior model uncertainty:

$$f(\mathcal{M}_\gamma \mid \mathbf{Y}) = \frac{f(\mathbf{Y} \mid \mathcal{M}_\gamma)f(\mathcal{M}_\gamma)}{\sum_{\mathcal{M}_\ell \in \mathcal{M}} f(\mathbf{Y} \mid \mathcal{M}_\ell)f(\mathcal{M}_\ell)} = \frac{f(\mathbf{Y} \mid \mathcal{M}_\gamma)f(\mathcal{M}_\gamma)}{C^{-1}}, \quad (6)$$

where $f(\mathbf{Y} \mid \mathcal{M}_\gamma)$ is called marginal likelihood:

$$f(\mathbf{Y} \mid \mathcal{M}_\gamma) = \int_{\Psi_\gamma} f(\mathbf{Y} \mid \boldsymbol{\psi}_\gamma, \mathcal{M}_\gamma)f(\boldsymbol{\psi}_\gamma \mid \mathcal{M}_\gamma) d\boldsymbol{\psi}_\gamma. \quad (7)$$

Bayesian Parametric Model Comparison

Each competing hypothesis is represented by a statistical model.

- **Null-Based Bayes Factor:** relative support $\mathbf{Y}$ provide for $\mathcal{M}_\ell$ over $\mathcal{M}_\emptyset$ (baseline):

$$\text{BF}_{\mathcal{M}_\ell:\mathcal{M}_\emptyset} = \frac{f(\mathbf{Y} | \mathcal{M}_\ell)}{f(\mathbf{Y} | \mathcal{M}_\emptyset)}. \quad (8)$$

- **Bayesian Ockham's razor effect Jefferys and J. O. Berger (1992):** among models consistent with the data, simpler models are naturally favored ($\Rightarrow$ sparsity). In (7), the likelihood is integrated over a higher-dimensional parameter space for more complex models, assigning less prior probability on the parameters that “actually” explain the data, making it more diffuse.
- **Implementation Challenges:** the specification of $f(\mathcal{M}_\ell)$ and $f(\boldsymbol{\psi}_\ell | \mathcal{M}_\ell)$; the evaluation of the integral in (7); and the computation of $f(\mathcal{M}_\ell | \mathbf{Y})$ (or $f(\boldsymbol{\ell} | \mathbf{Y})$) in high-dimensional settings.

Priors for Model Parameters

Subjective elicitation of genuine prior information is impractical $\Rightarrow$ using *default* prior distributions motivated by formal methods $\Rightarrow$ posterior information is driven primarily by the observed data.

Conventional Priors

- Computationally tractable and satisfy several desirable properties in a broad range of cases (Bayarri, J. O. Berger, et al. 2012; Li and Clyde 2018);
- Improper, non-informative priors to the common parameters, α and ϕ (if $\mathcal{M}_\emptyset$ is the baseline model) under certain conditions;
- Proper default priors to the remaining parameters (*basic criterion*; otherwise, indeterminate null-based BF);
- $f(\boldsymbol{\psi}_\gamma \mid \mathcal{M}_\gamma) = f(\boldsymbol{\beta}_\gamma \mid \phi, \mathcal{M}_\gamma) f(\alpha, \phi)$.

Zellner's g-prior

Null-based point-mass spike-and-slab prior (Zellner 1986):

$$\boldsymbol{\beta}_{\gamma} \mid \phi, \mathcal{M}_{\gamma} \sim \mathcal{N}_{k_{\gamma}}(\mathbf{0}, g \phi (\mathbf{X}_{\gamma}^{\top} \mathbf{X}_{\gamma})^{-1}). \quad (9)$$

- Only a single elicitation is required ($g > 0$);
- **Prior dispersion:** multiple of the variance of the ML Estimator of $\boldsymbol{\beta}_{\gamma}$, i.e., of the inverse of the expected Fisher information matrix (Dunn and Smyth 2018) $\Rightarrow$ model selection is scale-invariant (Hoff 2009);
- Conjugacy for normal likelihoods $\Rightarrow$ closed-form, rapid evaluation of (7).

Normal priors on $\boldsymbol{\beta}_{\gamma}$, however, are not conjugate to the likelihood, $f(\mathbf{Y} \mid \boldsymbol{\psi}_{\gamma}, \mathcal{M}_{\gamma})$, for non-normal EDMs $\Rightarrow$ no closed-form solution for the integral in (7) $\Rightarrow$ finding similar priors to (9) and use Laplace's method (Wang and George 2007).

Model Space Priors

Line of work: developing priors with desirable properties (e.g., adjustment for multiplicity), across a wide range of traditional model selection settings (interaction or higher-order terms; covariate redundancy; factors, etc.):

$$f(\boldsymbol{\gamma} \mid \mathbf{w}) = \prod_{j=1}^k f(\gamma_j \mid w_j) = \prod_{j=1}^k w_j^{\gamma_j} (1 - w_j)^{1-\gamma_j}, \quad (10)$$

where $\mathbf{w} = (w_1, \dots, w_k)$ and w_j denotes the prior probability of the j th variable being included in the model. Equivalently, $\gamma_j \mid w_j \sim \text{Ber}(w_j), j = 1, \dots, k$.

Advantages of a constant $w \Rightarrow$ model size $k_{\boldsymbol{\gamma}} \sim \text{Bin}(k, w) \Rightarrow w$ becomes the expected proportion of variables included in $\mathcal{M}_{\boldsymbol{\gamma}} \Rightarrow w = 0.5$ is the “noninformative” choice $\Rightarrow f(\boldsymbol{\gamma} \mid w = 0.5) = 2^{-k}$, the constant prior.

Multiple-Testing Problem

- Multiplicity correction arises from an appropriate choice of the space prior (Scott and J. O. Berger 2010), rather than inherently from the marginal likelihoods (i.e., it is distinct from the Bayesian Ockham's razor effect).
- "Pseudo-objective" choice $\Rightarrow$ prior odds favors models of intermediate dimension $\Rightarrow$ symmetric bell-shaped distribution on $k_{\boldsymbol{\gamma}}$ around $k/2$, with a more steep curve as k grows, regardless of $k_{\mathcal{M}_T} \Rightarrow$ more incorrect models, higher individual Type I error probability, and more false positives.
- **Empirical Bayes (EB)**: estimates w using ML (inducing dependence on k) $\Rightarrow$ controls for multiplicity, although the estimator may not infrequently yield degenerate solutions (e.g., assigning unwarranted certainty to the null model).
- **Fully Bayes (FB)**: places a prior on w , $\pi(w)$, usually a Beta(a,b), so that
$$f(\boldsymbol{\gamma}) = \int_0^1 f(\boldsymbol{\gamma} | w) \pi(w) dw,$$
 reducing prior weight of models of intermediate dimension.

FB solution to multiplicity adjustment

Scott and J. O. Berger (2010): $a = b = 1$, or, equivalently, $w \sim \text{Unif}(0, 1)$, leading to

$$f(\boldsymbol{\gamma}) = (k + 1)^{-1} \binom{k}{k_{\boldsymbol{\gamma}}}^{-1}. \quad (11)$$

- The same probability is assigned to each model dimension;
- **Prior probability of a model:** inversely proportional to the number of models of the corresponding dimension; viewed as a function of $k_{\boldsymbol{\gamma}}$, it follows a symmetric U-shaped curve, with peaks at model sizes 0 and k ;
- Models are penalized or favored depending on their position along the curve (multiplicity adjustment), which becomes increasingly steep as k increases;
- Same prior mean model size and individual prior inclusion probabilities as the “pseudo-objective” choice.

Posterior Distribution Over Models

Model selection decisions are based on $f(\boldsymbol{\gamma} \mid \mathbf{Y})$, which provides a complete representation of postdata model uncertainty. Below are two important deterministic functions of it:

- **Highest Posterior Model (HPM):** $\arg \max \{f(\boldsymbol{\gamma} \mid \mathbf{Y}) : \boldsymbol{\gamma} \in \mathcal{M}\}.$
- **Individual Posterior Inclusion Probabilities (PIPs):**

$$f(\gamma_j = 1 \mid \mathbf{Y}) = \sum_{\boldsymbol{\gamma} : \gamma_j = 1} f(\boldsymbol{\gamma} \mid \mathbf{Y}), \quad j = 1, \dots, k. \quad (12)$$

The model that includes all variables with PIPs exceeding the 1/2 threshold is the Median Probability Model (MPM) of Barbieri and J. O. Berger (2004); it exhibits strong predictive performance under a variety of settings.



High-Dimensional Settings

For high values of k , exact evaluation of (6) becomes infeasible (computing C involves a high computational burden).

Solution: sampling approaches that explore $\mathcal{M}$ by selecting a small subset of models, $\mathcal{M}^*$), relative to $|\mathcal{M}|$.

- **Renormalized approach:** heuristic procedures that involve (dependent) sampling without replacement; yield biased PIPs and exhibit high variability (García-Donato and Martínez-Beneito 2013);
- **Empirical approach:** sampling with replacement, with sampling probabilities equal to $f(\mathbf{Y} | \boldsymbol{\gamma}) f(\boldsymbol{\gamma}) \propto f(\boldsymbol{\gamma} | \mathbf{Y})$. Empirical frequencies are unbiased under independent draws; however, it is inadequate to specify all “size” variables $\Rightarrow$ obtain dependent samples.

Markov Chain Monte Carlo algorithms

- Simulate a sequentially dependent sample whose distribution converges to the stationary distribution, namely $f(\boldsymbol{\gamma} \mid \mathbf{Y})$:

$$\boldsymbol{\gamma}^{(0)}, \boldsymbol{\gamma}^{(1)}, \boldsymbol{\gamma}^{(2)}, \dots, \boldsymbol{\gamma}^{(S)} \quad (S \geq |\mathcal{M}^*|); \quad (13)$$

- $\boldsymbol{\gamma}^{(0)}$ influences only the rate of convergence $\rightarrow$ ergodic theorem (Hoff 2009);
- When the state space is finite, García-Donato and Martínez-Beneito (2013) show that the relative Monte Carlo empirical frequencies of the sampled models in $\mathcal{M}^*$, based on an MCMC sample drawn from $f(\boldsymbol{\gamma} \mid \mathbf{Y})$, are highly efficient and consistent ($\Rightarrow$ asymptotically unbiased as $S \rightarrow \infty$);
- HPM and PIPs are well approximated by their corresponding empirical counterparts; good ability to identify high-probable models;
- The sample size $|\mathcal{M}^*|$ needed to reach certain precision tends to finite values, even as $|\mathcal{M}|$ becomes arbitrarily large.

(Random-Order Componentwise) Gibbs Sampler

- Assumption of the presence of a finite state space, as in García-Donato and Martínez-Beneito (2013), although only closed-form expressions for an approximation of the marginal likelihood in (7) are derived.
- To generate the sequence in (13) it is easier to sample instead from the full conditional distributions of each γ_j (Hoff 2009):

$$f(\gamma_j | \mathbf{Y}, \boldsymbol{\gamma}_{(j)}) = \frac{f(\boldsymbol{\gamma} | \mathbf{Y})}{f(\boldsymbol{\gamma}_{(j)} | \mathbf{Y})} \propto f(\boldsymbol{\gamma} | \mathbf{Y}), \quad \boldsymbol{\gamma}_{(j)} = (\gamma_i : i \neq j). \quad (14)$$

- $\mathbb{E}[\gamma_j | \mathbf{Y}, \boldsymbol{\gamma}_{(j)}]$ is a function of the conditional posterior odds, o_j , that γ_j is 1:

$$o_j = \frac{f(\gamma_j = 1 | \mathbf{Y}, \boldsymbol{\gamma}_{(j)})}{f(\gamma_j = 0 | \mathbf{Y}, \boldsymbol{\gamma}_{(j)})} = \frac{f(\mathbf{Y} | \gamma_j = 1, \boldsymbol{\gamma}_{(j)})}{f(\mathbf{Y} | \gamma_j = 0, \boldsymbol{\gamma}_{(j)})} \times \frac{f(\gamma_j = 1, \boldsymbol{\gamma}_{(j)})}{f(\gamma_j = 0, \boldsymbol{\gamma}_{(j)})}. \quad (15)$$



03

Model Selection with Factors

Some Notes on Factors

- So far, only numerical variables were considered. Recall equation (5).
- Hereafter, $i \rightarrow$ numerical variables index; $j \rightarrow$ factors (categorical predictors); $t \rightarrow$ observations.
- Set of competing variables: $\mathbf{x}_1, \dots, \mathbf{x}_k$; and $\mathbf{\Lambda}_1, \dots, \mathbf{\Lambda}_p$, where $\ell_1, \dots, \ell_p$ are such that $\ell_j \geq 2, j = 1, \dots, p$, and $L = \sum_{j=1}^p \ell_j$.
- A numerical variable $\mathbf{x}_i$ is active iff $\beta_i \neq 0$.
- A factor $\mathbf{\Lambda}_j$ is active if at least one of the coefficients, $\lambda_{jr}, r = 1, \dots, \ell_j$, associated with its levels is non-zero, as in García-Donato and Paulo (2022).
- **Why this (flexible) definition?** E.g., factors from discretized numerical variables $\rightarrow$ their (somewhat arbitrary) level construction is subject to the complexity–fit trade-off (Dunn and Smyth 2018), so factor relevance depends on granularity (proportion of active levels to dimension) $\Rightarrow$ it is pertinent to consider models that allow for every possible combination of active levels within each factor.

Homoskedastic Linear Regression

$\mathcal{M}_F$ can be defined as:

$$\mathbf{y} = \mathbf{1}_n \alpha + \mathbf{X} \boldsymbol{\beta} + \mathbf{Z} \boldsymbol{\lambda} + \boldsymbol{\varepsilon}, \quad (16)$$

where the n -dimensional vector $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n)$.

- $\boldsymbol{\lambda} = (\boldsymbol{\lambda}_1, \dots, \boldsymbol{\lambda}_p)^\top$, with $\boldsymbol{\lambda}_j = (\lambda_{j1}, \dots, \lambda_{j\ell_j})^\top, j = 1, \dots, p$.
- $\mathbf{Z} = [\mathbf{Z}_1 \mid \dots \mid \mathbf{Z}_p]$ is a $n \times L$ matrix, where $\mathbf{Z}_j = (z_{tr}^{(j)})$ is an $n \times \ell_j$ matrix, with $z_{tr}^{(j)} = 1$ if the t -th observation takes the r th level of Λ_j , and 0 otherwise, $t = 1, \dots, n, r = 1, \dots, \ell_j, j = 1, \dots, p$.
- $z_r^{(j)}$: artificial or dummy variables (treatment coding).
- Indicator vector: $(\boldsymbol{\gamma}, \boldsymbol{\delta}) \in \{0, 1\}^{k+L}$, where $\boldsymbol{\delta}^\top = (\boldsymbol{\delta}_1^\top, \dots, \boldsymbol{\delta}_p^\top)$, $\boldsymbol{\delta}_j^\top = (\delta_{j1}, \dots, \delta_{j\ell_j})$; $\delta_{jr} = 1$ if level r of factor Λ_j is in the model (0 otherwise), $i = 1, \dots, k, r = 1, \dots, \ell_j, j = 1, \dots, p$.

Homoskedastic Linear Regression

- The matrix $\mathbf{Z}_{\boldsymbol{\delta}}$, of dimension $n \times p_{\boldsymbol{\delta}}$ (with $p_{\boldsymbol{\delta}} = \sum_{j=1}^p \sum_{r=1}^{\ell_j} \delta_{jr}$) is obtained by selecting the columns of $\mathbf{Z}$ indexed by $\boldsymbol{\delta}$; the corresponding subvectors of regression coefficients are denoted by $\boldsymbol{\lambda}_{\boldsymbol{\delta}}$.
- To indicate which factors are active in a particular model $\mathcal{M}_{\boldsymbol{\gamma}, \boldsymbol{\delta}}$, the indicator vector $\boldsymbol{\tau} = \boldsymbol{\tau}(\boldsymbol{\delta}) = (\tau_1(\boldsymbol{\delta}_1), \dots, \tau_p(\boldsymbol{\delta}_p))^{\top}$ is defined, where each τ_j is a deterministic function of $\boldsymbol{\delta}_j$, taking the value 1 if $\mathbf{1}_{\ell_j}^{\top} \boldsymbol{\delta}_j \geq 1$ (0 otherwise).

Overparameterized $\mathcal{M}_F$

Sometimes Bayes Factors are a function of ML estimates.

- ML estimation (Dunn and Smyth 2018) requires the design matrix $[\mathbf{1}_n \mid \mathbf{X}_\gamma \mid \mathbf{Z}_\delta]$ to be of full rank to ensure a unique solution exists.
- However, that will not always hold if $\mathcal{M}$ is generated by deleting columns from $[\mathbf{X} \mid \mathbf{Z}]$.
- To achieve this, under treatment coding, one can choose a reference level $r(j)$ for each factor Λ_j , setting $\lambda_{j,r(j)} = 0$ and omitting the corresponding column from $\mathbf{Z}_j$ to yield the reduced matrix $\mathbf{Z}_j^*$, for $j = 1, \dots, p$.
- On the other hand, generating $\mathcal{M}$ by deleting columns from $[\mathbf{X} \mid \mathbf{Z}^*]$, where $\mathbf{Z}^* = [\mathbf{Z}_1^* \mid \dots \mid \mathbf{Z}_p^*]$, does not yield an exhaustive model space, since not all possible combinations of active factor levels are represented (Fernández, Ley, and Steel 2002; García-Donato and Paulo 2022).

Exhaustive Model Space

The omitted models depend on the particular dummy encoding (reference level choice) $\Rightarrow$ what if $\mathcal{M}_T$ is *a priori* excluded from the candidate model set?

Deleting columns from $[\mathbf{X} \mid \mathbf{Z}]$ instead generates an exhaustive class of competing models (2^{k+L}). However, it contains duplicate models (different parameterizations of the same statistical model) $\rightarrow$ this occurs whenever, for at least one factor with ℓ levels, either $\ell - 1$ or all of its levels are active.

Solution from García-Donato and Paulo (2022):

- Remove all but one model from each set of duplicates. Which one?
- Retain only the overparameterized version. Why?
- It is simply a representative model, provided that the model evaluation criterion (e.g., the null-based BF in (8)) is invariant under full-rank parameterizations, true under conventional priors (Bayarri and García-Donato 2007).

Multiplicity Correction

Goal: prior indifference regarding the inclusion of a factor (e.g., Λ_j) irrespective of the set of potential explanatory variables under consideration, i.e.:

$$f(\tau_j = 1) = \sum_{\boldsymbol{\gamma}, \boldsymbol{\delta}: \tau_j = 1} f(\boldsymbol{\gamma}, \boldsymbol{\delta}) = \frac{1}{2}. \quad (17)$$

- Factors $\rightarrow$ grouped structures $\Rightarrow$ dependence in $(\boldsymbol{\gamma}, \boldsymbol{\delta}) \Rightarrow f(\tau_j = 1) \neq 1/2$ when $\ell_j > 2$ under both Constant and SB priors extensions to factors (García-Donato and Paulo 2022).
- Moreover, $f(\tau_j = 1)$ increases with ℓ_j and $f(\tau_j = 1)$ does not decrease as $k + p$ increases, regardless of the relevance of the additional levels or competing variables.

Two-stage Model Space Priors

Solution presented by García-Donato and Paulo (2022):

$$f(\boldsymbol{\gamma}, \boldsymbol{\delta}) = f(\boldsymbol{\gamma}, \boldsymbol{\delta}, \boldsymbol{\tau}) = f(\boldsymbol{\delta} \mid \boldsymbol{\gamma}, \boldsymbol{\tau})f(\boldsymbol{\gamma}, \boldsymbol{\tau}). \quad (18)$$

- The “marginal” prior $f(\boldsymbol{\gamma}, \boldsymbol{\tau})$ controls both $f(\gamma_i = 1)$ and $f(\tau_j = 1)$.
- The “conditional” prior $f(\boldsymbol{\delta} \mid \boldsymbol{\gamma}, \boldsymbol{\tau})$ distinguishes the probability assigned to models with the same $(\boldsymbol{\gamma}, \boldsymbol{\tau})$ combination, i.e., models belonging to the set $\mathcal{M}(\boldsymbol{\gamma}, \boldsymbol{\tau}) = \{(\boldsymbol{\gamma}, \boldsymbol{\delta}) \in \mathcal{M} : \boldsymbol{\gamma} = \boldsymbol{\gamma}, \min\{\mathbf{1}_{\ell_j}^\top \boldsymbol{\delta}_j, 1\} = \tau_j, j = 1, \dots, p\}$.
- **Examples:** constant and SB-type priors for both $f(\boldsymbol{\gamma}, \boldsymbol{\tau})$ and $f(\boldsymbol{\delta} \mid \boldsymbol{\gamma}, \boldsymbol{\tau})$.
- “Constant” marginals do not control for the multiplicity associated with $k + p$, regardless of $f(\boldsymbol{\delta} \mid \boldsymbol{\gamma}, \boldsymbol{\tau})$, unlike SB-type marginals.
- With an SB-type marginal, individual PIPs under both constant and SB-type conditional priors are sensitive to increases in ℓ_j .

Two-stage Model Space Priors: continuation

$\overline{B_{\gamma, \delta}^r}$: averaged BF, computed over $(\gamma, \delta) \in \mathcal{M}(\gamma, \tau)$ with the same model rank r .

- **SB-type conditional prior:** $\overline{B_{\gamma, \delta}^r}$ is a *simple* average of across all ranks in $\mathcal{M}(\gamma, \tau) \Rightarrow$ robust evidence for $\mathcal{M}(\gamma, \tau)$, regardless of $\mathcal{M}_T$'s rank.
- **Constant-type conditional prior:** $\overline{B_{\gamma, \delta}^r}$ is a *weighted* average, favoring intermediate ranks irrespective of $\mathcal{M}_T$'s rank.

Perks of this methodology:

- desiderata satisfied by conventional priors (Bayarri, J. O. Berger, et al. 2012)
- multiplicity control regarding $k + p$;
- no tuning parameters needed;
- comparably favourable frequentist performance to competing approaches.



$\Rightarrow$ is it possible to extend it to the broader class of GLMs?

Generalized Linear Models

Systematic Component

$$\boldsymbol{\eta}_{\gamma,\delta} = \mathbf{1}_n \alpha + \mathbf{X}_{\gamma} \boldsymbol{\beta}_{\gamma} + \mathbf{Z}_{\delta} \boldsymbol{\lambda}_{\delta} = \mathbf{1}_n \alpha + \tilde{\mathbf{X}}_{\gamma,\delta} \boldsymbol{\beta}_{\gamma,\delta} \quad (19)$$

- $\tilde{\mathbf{X}}_{\gamma,\delta} = [\mathbf{X}_{\gamma} \mid \mathbf{Z}_{\delta}]$ is the design matrix with $k_{\gamma,\delta} = k_{\gamma} + p_{\delta}$ columns and $\boldsymbol{\beta}_{\gamma,\delta} = (\boldsymbol{\beta}_{\gamma}, \boldsymbol{\lambda}_{\delta})^{\top}$.
- $\tilde{\mathbf{X}} = [\mathbf{X} \mid \mathbf{Z}]$ has full-column rank, with $k + p \leq n$, as its column space does not contain $\mathbf{1}_n \Rightarrow$ all categories of the factors are observed.
- $\mathcal{M}_T \in \mathcal{M}$; the ML estimates $\hat{\mathbf{c}} = (\hat{\alpha}_{\gamma,\delta}, \hat{\boldsymbol{\beta}}_{\gamma,\delta})$ are assumed to exist, are unique, and are consistent and asymptotically normal (Li and Clyde 2018).
- $\mathcal{J}_n(\hat{\mathbf{c}})$ is the observed Fisher information matrix, i.e., the negative Hessian matrix of the log-likelihood (Dunn and Smyth 2018), evaluated at the MLEs $\hat{\mathbf{c}}$
- Poisson, Binomial, and Exponential EDMs ($\phi = 1$).

Marginal Likelihood

Variants on Laplace's Method

$$f(\mathbf{Y} \mid \mathcal{M}_{\gamma, \delta}) = \int_{\mathbb{R}^{k_{\gamma} + p_{\delta}}} \int f(\mathbf{Y} \mid \alpha, \boldsymbol{\beta}_{\gamma, \delta}, \mathcal{M}_{\gamma, \delta}) f(\boldsymbol{\beta}_{\gamma, \delta} \mid \mathcal{M}_{\gamma, \delta}) f(\alpha) d\alpha d\boldsymbol{\beta}_{\gamma, \delta} \quad (20)$$

- The integrand $\propto f(\alpha, \boldsymbol{\beta}_{\gamma, \delta} \mid \mathbf{Y}, \mathcal{M}_{\gamma, \delta})$ is approximated by a normal distribution near its maximum (posterior mode).
- 2nd-order Taylor expansion of $\log f(\mathbf{Y} \mid \alpha, \boldsymbol{\beta}_{\gamma, \delta}, \mathcal{M}_{\gamma, \delta})$ around $\hat{\mathbf{c}} \Rightarrow$ likelihood $\approx$ p.d.f. of $\mathcal{N}(\hat{\mathbf{c}}, \mathcal{I}_n(\hat{\mathbf{c}})^{-1})$. Then, integrate, after substituting $f(\boldsymbol{\beta}_{\gamma, \delta} \mid \mathcal{M}_{\gamma, \delta}) f(\alpha) \approx f(\hat{\boldsymbol{\beta}}_{\gamma, \delta} \mid \mathcal{M}_{\gamma, \delta})$. See flaws in Wang and George (2007)
- **integrated Laplace (iL)**: same, but without the replacing the prior distribution;
- Same relative error, $O(n^{-1})$, for all; MLEs are easier to compute.

g-priors in GLMs

Normal g-priors, such as (9), require modification for non-normal EDMs, even when using the iL approximation.

These priors are commonly defined via a Fisher information matrix (either observed or expected), which depends on α and $\boldsymbol{\beta}_{\gamma,\delta}$ (r.v.'s) $\Rightarrow$ covariance matrix is not fixed, so the prior needs a rule for evaluating it:

- $\mathcal{I}_n(\boldsymbol{\beta}_{\gamma,\delta}) = \mathbb{E}[\mathcal{J}_n(\boldsymbol{\beta}_{\gamma,\delta})]$ (Fouskakis, Ntzoufras, and Draper 2009b) versus $\mathcal{J}_n(\boldsymbol{\beta}_{\gamma,\delta})$ (Wang and George 2007)? Efron and Hinkley (1978) recommend $\mathcal{J}_n(\boldsymbol{\beta}_{\gamma,\delta})$, as it more accurately measures the large-sample precision of $\hat{\boldsymbol{\beta}}_{\gamma,\delta}$;
- Where to evaluate it: at $\boldsymbol{\beta}_{\gamma,\delta} = \mathbf{0}$ or at $\hat{\boldsymbol{\beta}}_{\gamma,\delta}$? When $\mathcal{M}_T \neq \mathcal{M}_\emptyset$, evaluating at $\hat{\boldsymbol{\beta}}_{\gamma,\delta}$, the region where the likelihood concentrates, better captures the large-sample covariance and local likelihood geometry under $\mathcal{M}_{\gamma,\delta}$ (Wang and George 2007).

g-priors in GLMs: Li and Clyde (2018)

- Local Information Metric (LIM) g-prior:

$$\boldsymbol{\beta}_{\gamma, \delta} \mid \mathcal{M}_{\gamma, \delta}, g \sim \mathcal{N}(\mathbf{0}, g \cdot \mathcal{I}_n(\hat{\boldsymbol{\beta}}_{\gamma, \delta})^{-1}) \quad (21)$$

- Choice of g ?** Data-dependent values (Empirical Bayes approach on g) or adaptive values (e.g., $g = n$, as in unit information prior (Hoff 2009));
- EB is not model selection consistent when $\mathcal{M}_T = \mathcal{M}_\emptyset$;
- Mixtures of g-priors:** incorporate uncertainty in g through a hyper-prior. If the complementary shrinkage factor, $u = 1/(1 + g)$, follows a truncated Compound Confluent Hypergeometric (tCCH) distribution *a priori*, the result family, the Confluent Hypergeometric Information Criterion (CHIC), satisfies several desiderata, with the robust prior as a special case.

Prior on α

As hinted previously, α is assigned an improper prior of the type $f(\alpha) \propto 1$.

- Historically, that was justified by first recentering the columns of the design matrix $\tilde{\mathbf{X}}_{\gamma, \delta}$ to make them orthogonal to $\mathbf{1}_n$.
- However, Bayarri, J. O. Berger, et al. (2012) noted that this orthogonalization is unnecessary. Instead, they justify the improper prior through two key properties of the g-prior, both satisfied by the LIM g-prior in (21) (Li and Clyde 2018):
 - **Predictive Matching;**
 - **Location Invariance:** if integrating the marginal likelihood in (20) with respect to the improper uniform measure on α yields a marginal likelihood that is invariant to arbitrary location shifts in the columns of $\tilde{\mathbf{X}}_{\gamma, \delta} \Rightarrow$ no need to recenter.

Approximate Bayes Factor

Null-based BF under the **LIM** g-prior (Li and Clyde 2018), after an **iL** approximation:

$$\text{BF}_{\mathcal{M}_{\gamma,\delta}:\mathcal{M}_\emptyset} = \left[\frac{\mathcal{J}_n(\hat{\alpha}_{\mathcal{M}_\emptyset})}{\mathcal{J}_n(\hat{\alpha}_{\mathcal{M}_{\gamma,\delta}})} \right]^{\frac{1}{2}} (1+g)^{-\frac{k_{\gamma,\delta}}{2}} \exp \left\{ \frac{z_{\mathcal{M}_{\gamma,\delta}}}{2} - \frac{Q_{\mathcal{M}_{\gamma,\delta}}}{2(1+g)} \right\}. \quad (22)$$

- Wald test statistic under observed information, $Q_{\mathcal{M}} = \hat{\boldsymbol{\beta}}_{\gamma,\delta}^\top \mathcal{J}_n(\hat{\boldsymbol{\beta}}_{\gamma,\delta}) \hat{\boldsymbol{\beta}}_{\gamma,\delta}$;
- Deviance statistic (LLR) for $\mathcal{M}_{\gamma,\delta}$ vs $\mathcal{M}_\emptyset$: $z_{\mathcal{M}_{\gamma,\delta}} = 2 \log \left\{ \frac{f(\mathbf{Y}|\hat{\boldsymbol{\eta}}_{\mathcal{M}_{\gamma,\delta}}, \mathcal{M}_{\gamma,\delta})}{f(\mathbf{Y}|\hat{\boldsymbol{\eta}}_{\mathcal{M}_\emptyset}, \mathcal{M}_\emptyset)} \right\}$;
- Consistent with the Bayesian Ockham's razor principle (Jefferys and J. O. Berger 1992); model complexity penalty against goodness of fit reward (exponential term).
- Similar interpretation for the null-based BF under any mixture of g-priors member of the **CHIC** g-prior family (Li and Clyde 2018).

Bayes Factor Reference-Level Invariance

- The methodology of García-Donato and Paulo (2022) can be naturally extended to GLMs using the **LIM** g-prior, since (22) is unchanged by the choice of reference level for categorical variables in duplicate models.
- Detailed proof in **Appendix A**:
 - The conclusions also hold for the **CHIC** family of mixtures of g-priors.
 - Essentially, a change of reference level induces a linear transformation that leaves the linear predictor unaltered, but differs from the transformation underlying the measurement invariance property of both g-priors (Li and Clyde 2018).
 - This exercise also deepened my understanding of the mechanics of the *Fisher scoring* algorithm.

Simulation Experiments

Multiplicity correction properties for two-stage priors in GLMs

Goal: to examine the effect of different model prior specifications for $f(\boldsymbol{\tau})$ on multiplicity control, by investigating their impact on the posterior distribution, $f(\boldsymbol{\tau} \mid \mathbf{Y})$, as in García-Donato and Paulo (2022).

- Robust prior with the recommended hyperparameter values proposed by Bayarri et al. (2012). This choice is consistent with the recommendations from the empirical studies of Li and Clyde (2018).
- Poisson EDM with canonical link $\Rightarrow$ log-linear model (Dunn and Smyth 2018);
- Same design matrices as in García-Donato and Paulo (2022), but with different model-specific parameters. Multiplicative vs additive effect on the response means;
- $\alpha = 1$.

Experiment 1: Choice of Marginal-Stage Priors

What happens when spurious factors are added, holding the true model fixed?

Settings

- p factors, each with the same number of levels ($\ell = 4$). Three values for p were tested: $p \in \{5, 15, 27\}$. Each scenario was repeated 10 times, with a different design matrix and randomly chosen initial models for the Gibbs Samplers each time ($B = 2,000$, $S = 20,000$ and $|\mathcal{M}^*| \leq 5,000$).
- $\boldsymbol{\lambda}_1^\top = (-0.54, -0.42, -0.37, 0.315)$, $\boldsymbol{\lambda}_2^\top = (-0.255, 0.41, 0.18, 0.07)$, and $\boldsymbol{\lambda}_j = \mathbf{0}$, for $3 \leq j \leq p$.

Alert! Unambiguously defining active factors is not straightforward: a coefficient's effect depends on the model being tested (i.e., on which levels of that factor are included, and their values).

Experiment 1: Choice of Marginal-Stage Priors

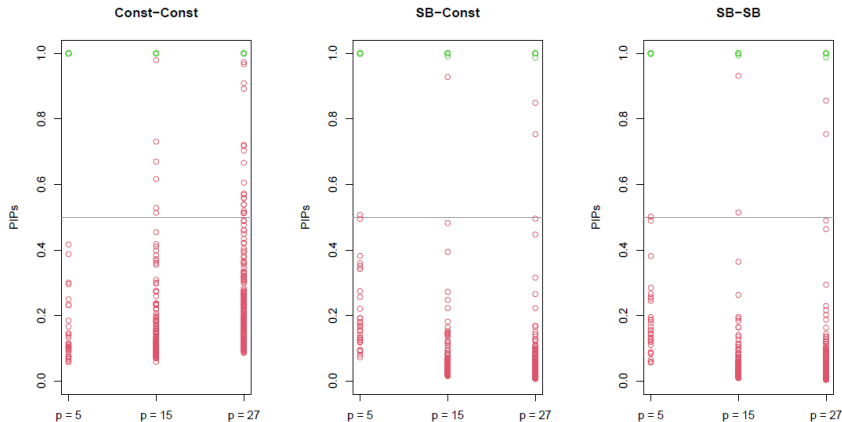


Figure 1: PIPs for Experiment 1 under the three prior specifications

Experiment 2: Comparing Conditional-Stage Priors

Only one active factor (out of $p = 4$), with a single active level, λ_{11} . No restriction on $|\mathcal{M}^*|$. The Gibbs samplers were run with $S = 100,000$ and $B = 10,000$.

What behavior is observed in PIPs as spurious categories increase, *ceteris paribus*?

	$\lambda_{11} = 0.70$				$\lambda_{11} = 0.55$			
	SBSB		SBC		SBSB		SBC	
	$\ell = 5$	$\ell = 15$	$\ell = 5$	$\ell = 15$	$\ell = 5$	$\ell = 15$	$\ell = 5$	$\ell = 15$
	$f(\tau_1 = 1 \mid \mathbf{Y})$	1.00	1.00	1.00	1.00	0.90	0.87	0.62
$f(\tau_2 = 1 \mid \mathbf{Y})$	0.15	0.08	0.09	0.00	0.07	0.08	0.03	0.00
$f(\tau_3 = 1 \mid \mathbf{Y})$	0.12	0.07	0.07	0.00	0.09	0.07	0.07	0.00
$f(\tau_4 = 1 \mid \mathbf{Y})$	0.07	0.07	0.03	0.00	0.11	0.08	0.02	0.00

Table 1: Mean PIPs across 10 simulation replicates in Experiment 2



04

Cost-Adjusted Model Selection

Motivation

- **Traditional VS Problem:** “benefit-only” analysis, with estimation or prediction as the sole goal.
- In real applications, predictors have associated **acquisition costs** (monetary, computational resources, data-collection time, modeling effort, etc.) $\Rightarrow$ the selection criterion should account for this additional information.
- Conflicting objectives:
 - **Joint optimization:** model the trade-offs explicitly (BMS $\leftrightarrow$ decision theory) via “cost–benefit” utility functions (Fouskakis and Draper 2008).
 - **Constrained optimization:** maximize predictive accuracy over the class of models satisfying a prespecified cost constraint (Fouskakis, Ntzoufras, and Draper 2009a).

Medical Applications

- Clinical decisions are resource-constrained: budgets set by hospitals or insurers;
- Selection must account for significant heterogeneous predictor costs while maintaining diagnostic accuracy;
- The “gold standard” predictor may be impractical (procedure too expensive or invasive) $\Rightarrow$ a cheaper subset with comparable performance is needed.
- **Gaps in existing approaches**
 - **Utility-based approaches:** preferences differ across stakeholders, so a common utility is hard to define.
 - **Cost-restrictive approaches:** budgets may be flexible, performance-dependent, or undefined *a priori*.
- $\Rightarrow$ need for a method in which practitioners control the degree of cost–performance trade-off through a single tuning parameter, according to the available budget and the desired model performance.



Cost-Sensitive Bayesian Model Selection

Fouskakis, Ntzoufras, and Draper (2009b)

- Individual data-collection costs are incorporated via the model space prior (**FND**)
⇒ models with different costs get different prior weights ⇒ costlier models are penalized *a priori*, unlike under standard VS.
- Inference as usual (HPM, MPM), now favoring less costly predictor subsets.
- Trading off cost against accuracy ⇒ a principled approximation to the utility-based approach, without specifying a utility function.
- **Important Property:** space prior is constructed so that the resulting posterior model odds (PO) generalize the **BIC**. Let $\mathcal{M}_{\boldsymbol{\gamma}}, \mathcal{M}_{\boldsymbol{\gamma}'} \in \mathcal{M}$ ($p = 0$). The PO of $\mathcal{M}_{\boldsymbol{\gamma}'}$ vs $\mathcal{M}_{\boldsymbol{\gamma}}$ can be defined as $\text{PO}_{\mathcal{M}_{\boldsymbol{\gamma}'}:\mathcal{M}_{\boldsymbol{\gamma}}} = f(\boldsymbol{\gamma}' | \mathbf{Y})/f(\boldsymbol{\gamma} | \mathbf{Y})$. Then

$$-2 \log \text{PO}_{\mathcal{M}_{\boldsymbol{\gamma}'}:\mathcal{M}_{\boldsymbol{\gamma}}} = -2 \log \text{BF}_{\mathcal{M}_{\boldsymbol{\gamma}'}:\mathcal{M}_{\boldsymbol{\gamma}}} - 2 \log \frac{f(\boldsymbol{\gamma}')}{f(\boldsymbol{\gamma})}. \quad (23)$$

FND Model Space Prior

Similar in spirit to the Bayesian Ockham's razor (built-in penalty for model complexity in the BF), the prior odds can provide a built-in penalty for costlier models by assigning cost-dependent penalties for individual predictors:

$$f(\boldsymbol{\gamma}) = \prod_{i=1}^k f(\gamma_i) = \prod_{i=1}^k \frac{\exp\left[-\frac{\gamma_i \log(n)}{2} \left(\frac{c_i}{c_0} - 1\right)\right]}{1 + \exp\left[-\frac{\log(n)}{2} \left(\frac{c_i}{c_0} - 1\right)\right]}, \quad (24)$$

where c_i is the cost of predictor i and $c_0 = \min\{c_i, i = 1, \dots, k\}$.

- $f(\gamma_i = 1)$ is a decreasing function of c_i (equal to 1/2 when $c_i = c_0$);
- Decisions invariant to cost-scale transformations (e.g., hours to euros using a prevailing wage rate).

Cost-Adjusted Generalized BIC

$\text{BIC}_{\mathcal{M}_{\gamma'}:\mathcal{M}_{\gamma}}$: fit (maximized likelihood) against complexity (model size) trade-off.
Replacing BF with BIC in (23) (biased PO, but no parameter priors needed):

$$-2 \log \text{PO}_{\mathcal{M}_{\gamma'}:\mathcal{M}_{\gamma}} \approx \text{BIC}_{\mathcal{M}_{\gamma'}:\mathcal{M}_{\gamma}} - 2 \log \frac{f(\gamma')}{f(\gamma)}. \quad (25)$$

Replacing (24) in (25), where $C_{\gamma} = \sum_{i=1}^k c_i \gamma_i$ is the cost of $\mathcal{M}_{\gamma}$:

$$-2 \log \text{PO}_{\mathcal{M}_{\gamma'}:\mathcal{M}_{\gamma}} \approx -2 \log \left[\frac{f(\mathbf{Y} \mid \hat{\alpha}_{\gamma'}, \hat{\beta}_{\gamma'}, \mathcal{M}_{\gamma'})}{f(\mathbf{Y} \mid \hat{\alpha}_{\gamma}, \hat{\beta}_{\gamma}, \mathcal{M}_{\gamma})} \right] + \frac{C_{\gamma'} - C_{\gamma}}{c_0} \log(n)$$

- Equal costs (constant prior) $\Rightarrow C_{\gamma}/c_0 = k_{\gamma}$ and PO have a BIC-like behavior.
- Different costs $\rightarrow$ scaled “complexity penalty” on the log-likelihood ratio by c_i/c_0 , relative to the BIC penalty $\log(n)$ at $c_i = c_0$, *ceteris paribus*.

Flexible Cost-Sensitive BIC

Under the **FND** prior, the rate of convergence of $f(\gamma_i = 1)$ to 0, as $n \rightarrow \infty$, may be inadequate for a given dataset or n .

Solution: adoption of Exponential Cost Penalizations (ECP), i.e., $c_i/c_0 \rightarrow (c_i/c_0)^b$, where b is a user-controlled tuning parameter (Porter, Franck, and Adams 2024).

Flexible FND prior properties:

- Cost conversion invariance is preserved; $\mathbf{b} = \mathbf{0} \rightarrow$ “benefit-only” analysis;
- Additional flexibility $\Rightarrow$ multiple cost-adjusted analysis over a range of b values $\Rightarrow$ practitioners can choose the cost penalization that best suits their desired cost-performance trade-off;
- Prior IPs are a decreasing function of b , *ceteris paribus*;
- $\mathbf{b} \neq \mathbf{1} \rightarrow$ posterior odds no longer **BIC**-like, but still cost-sensitive.

Multiple-testing Problem

The **Flexible FND prior** does not control for multiplicity when all costs are equal (it generalizes a constant-type marginal, not an SB-type marginal). Addressing this fell outside the scope of this work, since:

- any solution would depend on the particular cost structure;
- a prior inversely proportional to the number of models of the same cost is not well motivated, and it is unclear how to define it;
- data-collection costs reflect predictor-specific external information, not aggregate information (e.g., model size or number of models of that size).

⇒ **Limitation!** Be alert to baseline predictors with uncertain effects, as their prior IP is not affected by the cost penalization.

Cost-Adjusted Model Selection with Factors

- The methodology in Porter, Franck, and Adams (2024) and Fouskakis, Ntzoufras, and Draper (2009b) applies only when factors are encoded as ordinal ($\ell > 2$) or binary ($\ell = 2$) variables, since these have a single associated coefficient.
- The dissertation identifies situations where ordinal encoding is not appropriate (e.g., arbitrary scores) $\Rightarrow$ motivating an extension of the methodology in García-Donato and Paulo (2022) to this setting.
- The data-collection cost should be associated with the predictor itself, rather than with the value it takes (Fouskakis, Ntzoufras, and Draper 2009b) $\Rightarrow$ cost penalization is set at the group level, not at the dummy level.

Flexible FND prior with Factors

$$f(\boldsymbol{\gamma}, \boldsymbol{\tau}) \propto \prod_{i=1}^k \exp \left[-\frac{\gamma_i \log(n)}{2} \left(\frac{c_i}{c_0} - 1 \right) \right] \prod_{j=1}^p \exp \left[-\frac{\tau_j \log(n)}{2} \left(\frac{c_j}{c_0} - 1 \right) \right] \quad (26)$$

The baseline cost is now $c_0 = \min \left(\{c_i\}_{i=1}^k \cup \{c_p\}_{j=1}^p \right)$ and the posterior odds take now the form:

$$-2 \log \text{PO}_{\mathcal{M}_{\boldsymbol{\gamma}', \boldsymbol{\delta}'} : \mathcal{M}_{\boldsymbol{\gamma}, \boldsymbol{\delta}}} = \text{BIC}_{\mathcal{M}_{\boldsymbol{\gamma}', \boldsymbol{\delta}'} : \mathcal{M}_{\boldsymbol{\gamma}, \boldsymbol{\delta}}} - 2 \log \frac{f(\boldsymbol{\gamma}', \boldsymbol{\tau}')}{f(\boldsymbol{\gamma}, \boldsymbol{\tau})} - 2 \log \frac{f(\boldsymbol{\delta}' | \boldsymbol{\gamma}', \boldsymbol{\tau}')}{f(\boldsymbol{\delta} | \boldsymbol{\gamma}, \boldsymbol{\tau})} + O(1)$$

Interestingly, when $b = 0$, the posterior odds no longer display **BIC**-like behavior under either of the conditional prior proposals in García-Donato and Paulo (2022).

Conditional Prior Odds

$\frac{f(\boldsymbol{\delta}'|\boldsymbol{\gamma}',\boldsymbol{\tau}')}{f(\boldsymbol{\delta}|\boldsymbol{\gamma},\boldsymbol{\tau})}$ is not guaranteed to be 1 for any pair of models in $\mathcal{M}$:

- **SB-type conditional:** e.g., for two same-cost models with $\mathcal{M}(\boldsymbol{\gamma}', \boldsymbol{\tau}') = \mathcal{M}(\boldsymbol{\gamma}, \boldsymbol{\tau})$, the prior odds may now favor higher-rank models, unlike under the **FND** prior with numerical variables only.
- **Constant-type conditional:** e.g., consider two models with $\boldsymbol{\gamma}' = \boldsymbol{\gamma}$, but $\sum_{j=1}^p (\tau'_j - \tau_j) = 1$ and $k_{\boldsymbol{\gamma}', \boldsymbol{\delta}'} = k_{\boldsymbol{\gamma}, \boldsymbol{\delta}} - 1$, with the extra active factor satisfying $l_j \geq 3 \Rightarrow |\mathcal{M}(\boldsymbol{\gamma}', \boldsymbol{\tau}')| > |\mathcal{M}(\boldsymbol{\gamma}, \boldsymbol{\tau})| \Rightarrow$ the overall penalty for adding a single level of the extra factor is higher than under the **FND** prior with numerical variables only.

Still, what are the implications of each option for cost-adjusted model selection?

Cost Structures

In some situations, a predictor's data-collection cost depends on whether other predictors are collected simultaneously $\Rightarrow$ the independence assumption no longer holds:

- **Grouped predictors:** a single procedure yields several predictors at no extra cost (e.g., multiple biomarkers from one blood sample);
- **Discounted predictors:** predictors that become cheaper to collect jointly, owing to shared equipment, personnel, or preparation.

$\Rightarrow$ the current cost-adjusted prior needs adjusting, as it may overestimate the true data-collection burden.

For ease of exposition, it is assumed: only numerical variables; a single cost structure, $\mathbf{A} = \{\mathbf{x}_1, \dots, \mathbf{x}_{|A|}\}$, with $2 \leq |A| \leq k$; no change in c_0 ; and two-stage cost dependencies.



Cost Structures: Notation

$c_i \rightarrow$ marginal / full cost; $c_i^* \rightarrow$ discounted cost (with $c_0 < c_i^* < c_i$):

- **Grouped predictors:** $c_i \rightarrow$ procedure cost (single variable included); $c_i^* = c_0 = 1$ (provided that another predictor from the same group is active);
- **Discounted predictors:** $c_i \rightarrow$ cost when no other variable of the group is included ("first"); $c_i^* \rightarrow$ cost provided that another predictor from the same group is active ("second" or "subsequent");

Two-stage cost dependencies: $c_i + \sum_{s \in A \setminus \{i\}} c_s^*$ takes the same value for all i in $\mathbf{A}$; otherwise, redefine discounted costs $\rightarrow$ offline decision-making.

$\Rightarrow$ New model space prior ($f(\gamma_i)$ follows from the original **Flexible FND prior**):

$$f(\boldsymbol{\gamma}) = f_{\mathbf{A}}(\gamma_1, \dots, \gamma_{|\mathbf{A}|}) \prod_{i=|\mathbf{A}|+1}^k f(\gamma_i). \quad (27)$$

Cost-Adjusted Model Prior with Cost Structures

$$f_{\mathbf{A}}(\gamma_1, \dots, \gamma_{|\mathbf{A}|}) \propto \max_{i \in \mathbf{A}} \left\{ f^*(\gamma_i) \cdot \prod_{\substack{s \in \mathbf{A} \\ s \neq i}} f_d^*(\gamma_s \mid \gamma_i) \right\}, \quad (28)$$

where

$$\begin{aligned} f^*(\gamma_i) &= \exp \left\{ -\frac{\gamma_i \log(n)}{2} \left(\left(\frac{c_i}{c_0} \right)^b - 1 \right) \right\}, \\ f_d^*(\gamma_s \mid \gamma_i) &= \exp \left\{ -\frac{\gamma_s \log(n)}{2} \left(\left(\frac{c_s(1 - \gamma_i) + c_s^* \gamma_i}{c_0} \right)^b - 1 \right) \right\}. \end{aligned} \quad (29)$$

- Neither $f^*(\gamma_i)$ nor $f_d^*(\gamma_s \mid \gamma_i)$ are probability distributions.
- No need to compute the normalizing constant (no effect on prior odds).



Cost-Adjusted Model Prior with Cost Structures

Properties

- If $c_s = c_s^*$, $\forall s \in \mathbf{A} \Rightarrow f_d^*(\gamma_s \mid \gamma_i) = f^*(\gamma_i) \Rightarrow f(\boldsymbol{\gamma}) = \prod_{i=1}^k f(\gamma_i)$.
- Posterior Odds generalize **BIC** when $b = 1$ (only numerical variables).
- Can it be shown that $f_{\mathbf{A}}(\gamma_i = 1) \leq 1/2$?
- What is the relation between $f_{\mathbf{A}}(\gamma_i = 1)$ and $f(\gamma_i = 1)$? Contradicting effects, but which is stronger?
 - $f_{\mathbf{A}}(\gamma_i = 1, \sum_{s \in \mathbf{A}} \gamma_s = 1) < f(\gamma_i = 1, \sum_{s \in \mathbf{A}} \gamma_s = 1) \Rightarrow$ negative effect in $f_{\mathbf{A}}(\gamma_i = 1)$ in comparison to $f(\gamma_i = 1)$;
 - $f_{\mathbf{A}}(\gamma_i = 0, \sum_{s \in \mathbf{A}} \gamma_s \geq 1) < f(\gamma_i = 0, \sum_{s \in \mathbf{A}} \gamma_s \geq 1) \Rightarrow$ several situations with a positive effect in $f_{\mathbf{A}}(\gamma_i = 1)$ in comparison to $f(\gamma_i = 1)$.

Case Study: Logistic Regression

Goal: evaluate the proposed approaches for incorporating factors and cost structures into cost-penalized Bayesian model selection.

- Real dataset of clinical test results from the Cleveland Clinic Foundation. The target variable indicates whether a patient has heart disease. More details on the candidate predictors can be found in Porter, Franck, and Adams (2024);
- In-sample predictive performance is reported and measured by the Area Under the Receiver Operating Characteristic (ROC) curve, also known as the Concordance (C) statistic (Porter, Franck, and Adams 2024).
- Random-order componentwise Gibbs samplers were run with $S = 100,000$, $B = 10,000$, and random initial models.

Case Study: Logistic Regression

- **LIM** g-prior in (21) with $g = n$, due to its close relationship with the **BIC**.
- Using large sample asymptotic arguments (Li and Clyde 2018), posterior modes for model parameters are functions of the ML estimates:

$$\begin{aligned}\boldsymbol{\beta}_{\gamma,\delta} \mid \mathbf{Y}, \mathcal{M}, g &\xrightarrow{D} \mathcal{N}\left(\frac{g}{1+g}\hat{\boldsymbol{\beta}}_{\gamma,\delta}, \frac{g}{1+g}\mathcal{J}_n(\hat{\boldsymbol{\beta}}_{\gamma,\delta})^{-1}\right) \\ \alpha \mid \mathbf{Y}, \mathcal{M} &\xrightarrow{D} \mathcal{N}(\hat{\alpha}_{\gamma,\delta}, \mathcal{J}_n(\hat{\alpha}_{\gamma,\delta})^{-1})\end{aligned}\tag{30}$$

- The marginal posterior distribution of α is asymptotically proper and independent of that of $\boldsymbol{\beta}_{\gamma,\delta}$ (local orthogonality between).
- $\Rightarrow$ efficient computation of the fitted values under both the HPM and the MPM, facilitating the evaluation of their corresponding C-statistics.
- Posterior modes = Posterior means = Posterior medians.

Dataset Information

Variable	Description	Marginal Cost	Discounted Cost	Group
age	Patient age	1.00	–	–
sex	Patient sex	1.00	–	–
chest pain	Chest pain type	1.00	–	–
resting BP	Resting blood pressure	1.00	–	–
cholesterol	Serum cholesterol	7.27	5.17	A
blood sugar	Fasting blood sugar	5.20	3.10	A
EKG	Resting ECG results	15.50	–	–
heart rate	Maximum heart rate achieved	102.90	1.00	B
exercise angina	Exercise-induced angina	87.30	1.00	C
ST depression	Exercise-induced ST depression vs. rest	87.30	1.00	C
peak ST segment	Slope of peak exercise ST segment	87.30	1.00	C
major vessels	Major vessels colored by fluoroscopy	100.90	–	–
defect type	Type of heart defect	102.90	1.00	B

Table 2: Acquisition costs in Canadian dollars. Variables with no group discount are marked with –.

Case-Study

Cost Structures:

- **Group A:** two distinct procedures that clearly benefit from time savings when performed one “after” the other.
- **Group B** → Thallium stress test: nuclear imaging test (myocardial perfusion imaging) that shows how well blood flows through the heart muscle, both during stress and at rest.
- **Group C** → Exercise stress test procedure: the patient walks or runs on a treadmill with progressively increasing intensity while an electrocardiogram (ECG) records the heart’s electrical activity, along with blood pressure and symptoms.

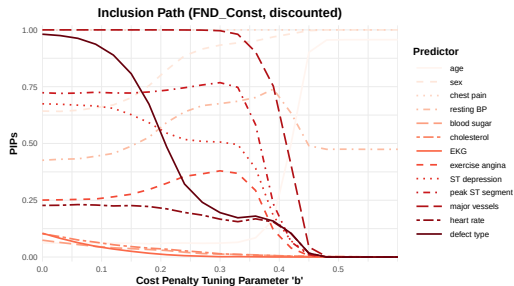
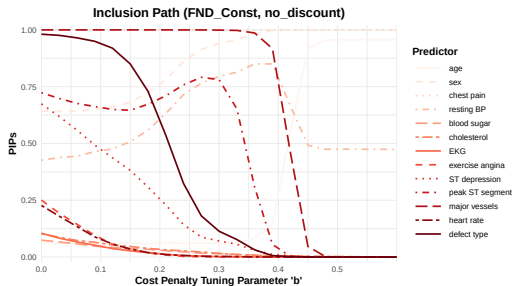
Inclusion Paths: Twenty-one cost-adjusted analyses were performed for each proposal, with b ranging from 0 to 0.6 in increments of 0.03. No changes in the results were observed for values of b above 0.6.



FND-type Marginal & Constant Conditional

Inclusion Paths

Same results for $b = 0$, obviously (benefit-only analysis).

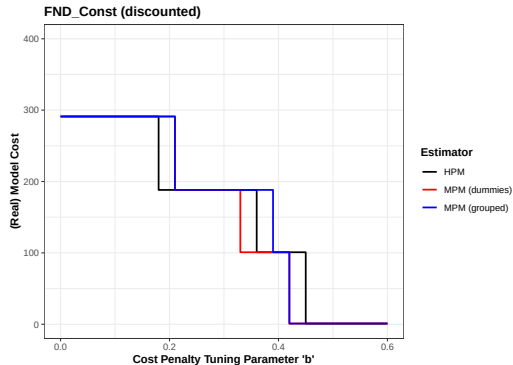
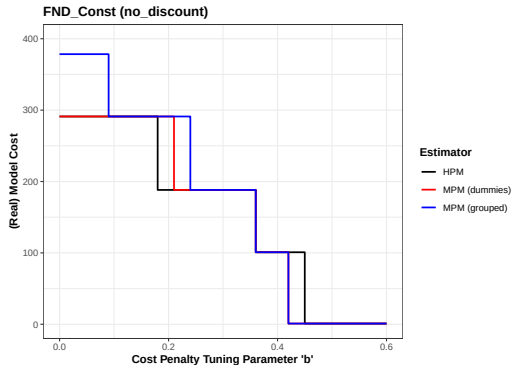


Left: No Discounts; Right: With Discounts.

FND-type Marginal & Constant Conditional

Model Cost

Some differences reflect genuinely different selected models; others merely stem from how predictor costs are defined.



Left: No Discounts; Right: With Discounts.

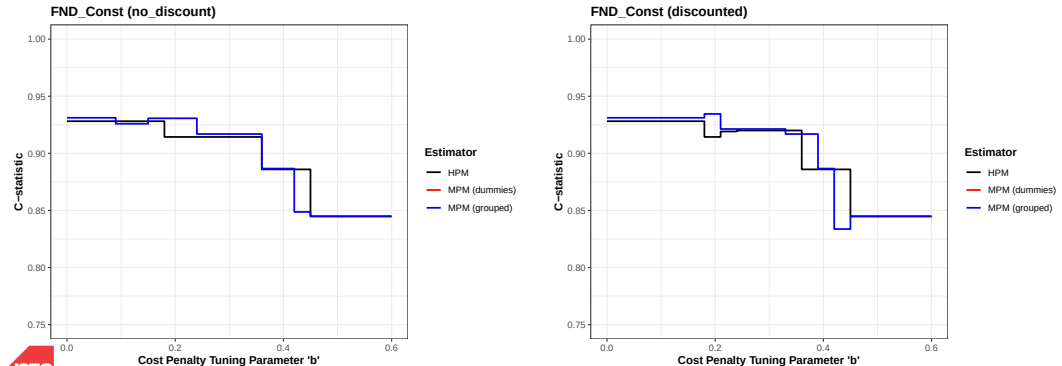
Median Probability Model with Factors

- **MPM dummies:** all numerical predictors with PIPs > 0.5 , plus only the dummy variables whose PIPs also exceed 0.5.
- **MPM grouped:** same as **MPM dummies** for numerical variables, but all levels of a factor are included whenever the factor's PIP exceeds 0.5 $\rightarrow$ the prior IP is fixed at the factor level (0.5), not at the dummy level.
- **MPM grouped** is at least as large as **MPM dummies** $\Rightarrow$ sometimes different model costs.
- Even at identical costs, their relative usefulness depends on the application (Exp. 1 vs Exp. 2 settings) $\rightarrow$ discrepancies may indicate that some levels contribute little to prediction, suggesting they could be merged (García-Donato and Paulo 2022).

FND-type Marginal & Constant Conditional

C-statistic

Substantial cost reductions at only minor predictive losses. Performance comparable to Porter, Franck, and Adams (2024) despite different treatment of factors $\Rightarrow$ appropriate categorization of ordinal predictors (e.g., defect type: 3, 6, 7).

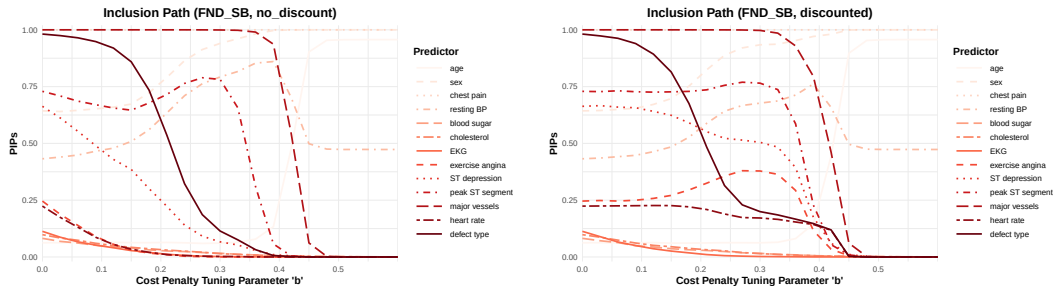


Left: No Discounts; Right: With Discounts.

FND-type Marginal & SB-type Conditional

Inclusion Paths

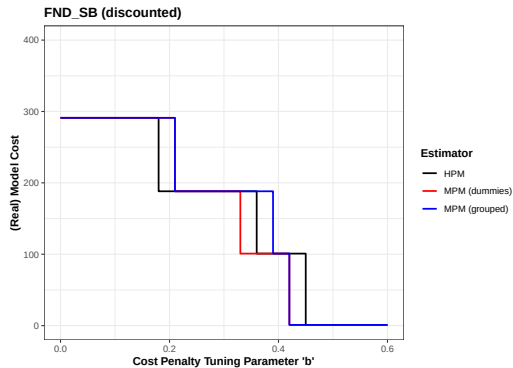
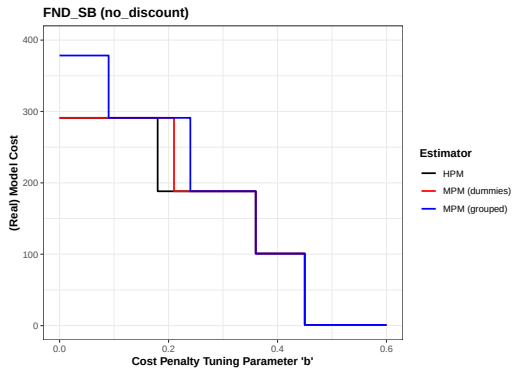
As expected, the two conditional priors yielded only marginally different results, since the factors have at most $\ell = 4$ levels.



Left: No Discounts; Right: With Discounts.

FND-type Marginal & SB-type Conditional

Model Cost



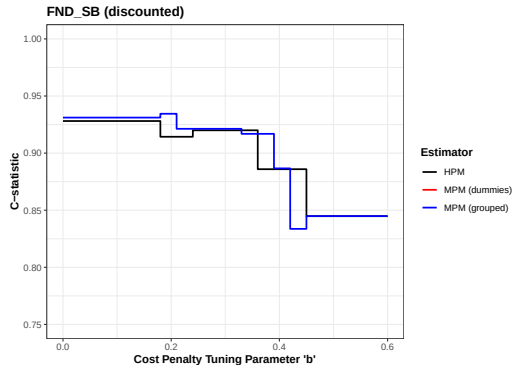
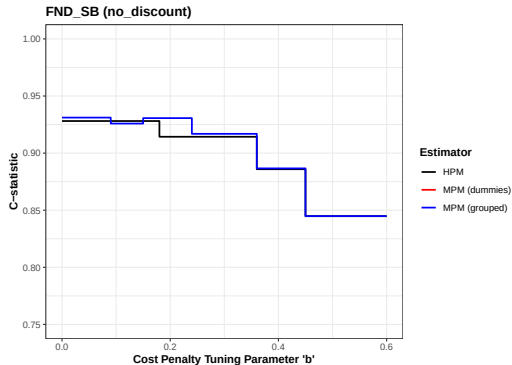
Left: No Discounts; Right: With Discounts.



FND-type Marginal & SB-type Conditional

C-statistic

The discounted approach does not consistently outperform the standard cost specification in terms of predictive accuracy across all values of b .



Left: No Discounts; Right: With Discounts.



05

Conclusions

Key Takeaways

- LIM and CHIC g-priors approximate BF's (Li and Clyde 2018) are invariant to reference-level choice $\Rightarrow$ extension of methodology in García-Donato and Paulo (2022) to GLMs.
- Extended cost-adjusted Bayesian Model Selection to factors $\Rightarrow$ cost penalties connected with $(\boldsymbol{\gamma}, \boldsymbol{\tau})$ instead of $(\boldsymbol{\gamma}, \boldsymbol{\delta})$.
- Proposed prior accommodating non-independent data-collection costs for grouped/discounted predictors, although discounted specification did not uniformly dominate its independent-cost counterpart for the Cleveland data.

Difficulties in Designing Simulation Studies

Evidence between conditional priors too volatile to be conclusive. Poor identification of active factors for other EDMs at $n = 300$.

- **Laplace approximation:** error order $O(n^{-1})$ is taken to be EDM-independent, but approximation quality at fixed n depends on the EDM's geometry (Slate 1994).
- **Simulation design itself:**
 - Link function $\neq$ identity $\Rightarrow$ active factor detectability depends on separation in unit deviance (asymmetric and non-standard for non-Normal EDMs), not raw response scale.
 - (Total) deviances are not comparable across EDMs $\Rightarrow$ calibration of model parameters is not straightforward (e.g., choice of intercept now matters, due to mean-dependent variance functions).

Future Research

- **Gibbs Sampler:** full conditionals ignore whether current/proposed models lie in the real $\mathcal{M} \Rightarrow$ explored model space contains repeated models $\Rightarrow$ post hoc weighted resampling. Biased estimator of posterior model probabilities? How to implement a search restricted to the valid $\mathcal{M}$ directly?
- **Sequential Decision-Making:** in real-world medical diagnosis, tests are ordered sequentially based on patient results. E.g., how could the methodology in Ji and Carin (2007), which relies on Hidden Markov models (cheap tests first, update beliefs, order costlier tests only while justified), be adapted to this work's cost structures and flexible cost penalization?
- **Dilution Priors:** Grouped prior inclusion probability could be controlled, similar in spirit to Chipman (1996), as grouped predictors (same procedure) may carry overlapping information.



Thanks for your attention!

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David Baptista

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